

Azwad R. Iqbal

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Professional Summary

I am a broadly trained population geneticist and evolutionary biologist interested in rapid evolution and conservation. My research leverages cutting-edge genomic techniques, bioinformatics, and fieldwork to understand the evolutionary consequences of anthropogenic change.

Education

Cornell University

PH.D. NATURAL RESOURCES AND THE ENVIRONMENT

- Advisor: Nina Overgaard Therkildsen

Ithaca, NY

2021 - 2027 (expected)

Princeton University

B.A. ECOLOGY AND EVOLUTIONARY BIOLOGY

- *Summa Cum Laude*
- Minor in Environmental Studies

Princeton, NJ

2015 - 2019

Research Experience

McBride Lab, Princeton University

RESEARCH SPECIALIST

Princeton, NJ

May 2019 - July 2021

Pringle Lab, Princeton University

SENIOR THESIS RESEARCH

Princeton, NJ

June 2018 - April 2019

Ayroles Lab, Lewis-Sigler Institute for Integrative Genomics

LAB TECHNICIAN

Princeton, NJ

January - December 2018

Pringle Lab, Nyika National Park

FIELD TECHNICIAN

Malawi

June - August 2017

Pringle Lab, Mpala Research Centre

FIELD TECHNICIAN

Laikipia, Kenya

July - September 2016

Publications

1. **Iqbal, A. R.**, Dimens, P. V., Rick, J. A., Munn, P. R., McNairn, A. J., Landis, J. B., Schembri, R., Chan, Y. F., Kučka, M., Therkildsen, N. O., & Grenier, J. K. (2026). *BLink-seq delivers population-scale haplotypes without long reads: A scalable framework for non-model genomics*. <https://doi.org/10.64898/2026.08.03.742036>
2. Velotta, [†] J. P., **Iqbal, [†] A. R.**, Glenn, E. S., Franckowiak, R. P., Formenti, G., Mountcastle, J., Balacco, J., Tracey, A., Sims, Y., Howe, K., Fedrigo, O., Jarvis, E. D., & Therkildsen, N. O. (2024). A complete assembly and annotation of the american shad genome yields insights into the origins of diadromy. *Genome Biology and Evolution*, 17(1). <https://doi.org/10.1093/gbe/evae276>
3. Dimens, P. V., Franckowiak, R. P., **Iqbal, A.**, Grenier, J. K., Munn, P. R., & Therkildsen, N. O. (2024). Harpy: A pipeline for processing haplotagging linked-read data. *Bioinformatics Advances*, 5(1). <https://doi.org/10.1093/bioadv/vbaf133>
4. Pansu, J., Hutchinson, M. C., Anderson, T. M., Beest, M. te, Begg, C. M., Begg, K. S., Bonin, A., Chama, L., Chamailé-Jammes, S., Coissac, E., Cromsigt, J. P. G. M., Demmel, M. Y., Donaldson, J. E., Guyton, J. A., Hansen, C. B., Imakando, C. I., **Iqbal, A.**, Kalima, D. F., Kerley, G. I. H., ... Pringle, R. M. (2022). The generality of cryptic dietary niche differences in diverse large-herbivore assemblages. *Proceedings of the National Academy of Sciences*, 119(35). <https://doi.org/10.1073/pnas.2204400119>
5. Zhao, Z., Zung, J. L., Hinze, A., Kriete, A. L., **Iqbal, A.**, Younger, M. A., Matthews, B. J., Merhof, D., Thiberge, S., Ignell, R., Strauch, M., & McBride, C. S. (2022). Mosquito brains encode unique features of human odour to drive host seeking. *Nature*, 605(7911), 706–712. <https://doi.org/10.1038/s41586-022-04675-4>

Presentations

- 2026 Founder effects, chromosomal inversions, and selective sweeps shape the rapid evolution of an invasive fish. *Evolution 2026, Cleveland, OH*
- 2026 Founder effects, local adaptation, and chromosomal inversions shape the rapid evolution of an invasive migratory fish (Poster). *PEQG 2026, Monterey, CA*
- 2024 Founder effects, local adaptation, and chromosomal inversions shape the rapid evolution of invasive American shad. *Evolution 2024, Montreal, QB, Canada*
- 2023 Insights into the Pacific Coast Invasion of American shad (Poster). *Great Lakes Annual Meeting of Evolutionary Genomics, Ithaca, NY*
- 2022 Investigating the genomic basis of rapid adaptation in the American Shad (Poster). *American Fisheries Society Annual Meeting, Spokane, WA*
- 2022 Investigating the genomic basis of rapid adaptation in an invasive fish. *Evolution 2022, Cleveland, OH*

Funding & Grants

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| 2025 | Cornell Atkinson Center Sustainable Biodiversity Fund | \$2,970 |
| 2025 | American Museum of Natural History Theodore Roosevelt Memorial Fund Grant | \$2,400 |
| 2024 | Cornell Center for Vertebrate Genomics Scholar Award | \$12,000 |
| 2023 | American Philosophical Society Lewis and Clark Fund | \$3,000 |
| 2023 | Society for the Study of Evolution (SSE) R.C. Lewontin Early Award | \$2,500 |
| 2023 | National Science Foundation Graduate Research Fellowship (GRFP) | \$159,000 |
| 2021 | Cornell University Fellowship | \$33,032 |
| 2021 | Cornell Atkinson Center Sustainable Biodiversity Fund | \$6,930 |

Honors & Awards

- 2024 Cornell Center for Vertebrate Genomics Scholar
- 2022 Best Proposal-stage Presentation - Cornell Department of Natural Resources Symposium
- 2021 NSF Graduate Research Fellowship Program (GRFP) Honorable Mention
- 2019 Sigma Xi Book Award for Outstanding Research - Princeton Department of Ecology & Evolutionary Biology
- 2019 Senior Thesis Prize in Ecology - Princeton Department of Ecology & Evolutionary Biology
- 2019 Gates Cambridge Finalist - Biological Sciences

Teaching

- 2025-26 *Guest Lecture* - NTRES 2830: DNA, Genes and Genetic Diversity
- 2023, 26 *Teaching Assistant* - NTRES 6100: Collaborative and Reproducible Data Science in R
- 2022 *Teaching Assistant* - NTRES 2100: Introductory Field Biology
- 2021 *Guest Lecture* - NTRES 3400: Molecular Tools for Ecology, Conservation, and Natural Resource Management
- 2019 *Co-Instructor* - St. Paul's School Advanced Studies Program (Ecology)

Leadership & Service

- 2022-24 *Co-facilitator* - Cornell EvoGroup Seminar Series
- 2023 *Volunteer* - Cornell Expand Your Horizons
- 2022-23 *General Council Member* - Cornell Department of Natural Resources DEI Advisory Council
- 2022 *Co-Instructor* - Cornell Graduate Student School Outreach Program (GRASSHOPR)

Training & Workshops

- 2025 Invasion Genomics in R - Physalia
- 2025 Machine and Deep Learning methods in Population Genomics and Phylogeography - Physalia
- 2024 Introduction to ancient DNA data analysis - American Museum of Natural History
- 2023 SLiM Workshop - Cambridge, MA
- 2022 Incorporating paleogenomes into evolutionary genomics studies - Physalia
- 2021 Population genomic inference from low-coverage whole-genome sequencing data - Physalia

Skills

Computational: R, Python, Snakemake, bash, SLURM, bwa, samtools/bcftools, ANGSD, GATK, git/GitHub

Laboratory: Gel electrophoresis, PCR, gDNA extraction (archival & contemporary tissues), NGS library preparation, BLink-seq